

Theoretically Analyses the Role of Microcavities in Adjusting the Light Extraction Efficiency of QLEDs

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Abstract—We developed an innovative simulation method using Leda software to optimize light extraction in QLEDs. By simply adjusting the thickness of the semitransparent metal layer, we achieved nearly a 100% improvement in light extraction efficiency (LEE), raising it from 15.88% to 31.16% under optimal conditions. This remarkable approach holds great potential for broader applications in QLED devices.

Keywords—Leda; Light extraction efficiency; QLEDs; Microcavities

I. INTRODUCTION

Quantum dot light-emitting diodes (QLEDs) are characterized by high luminous efficiency, tunable emission colors, and narrow emission peaks^[1-3]. They are a key direction for next-generation display technology^[4,5], particularly due to their solution-processable characteristics, which show unique advantages in flexible display applications. Recent advancements in colloidal quantum dot (QDs) synthesis, such as gradient alloying and shell engineering, have further improved the photoluminescence quantum yield (PLQY) of QDs to over 95%. However, the electroluminescence efficiency of QLEDs remains constrained by optical losses,

including waveguide modes and substrate trapping. Addressing these losses through photonic structure design is thus imperative for bridging the gap between laboratory prototypes and commercial viability. To meet the demands of cutting-edge display technologies, particularly the development of QDs LED devices, it is crucial to break through the limitations of transport layer materials in electroluminescent devices and develop novel electron transport materials with originality and independent intellectual property rights. Currently, the electron transport layer (ETL) in QLEDs mainly consists of ZnO or ZnMgO nanocrystals^[4,6-8]. Although it offers relatively high performance, the surface modification of ZnO at the nanoscale remains challenging, primarily stemming from the high reactivity of surface hydroxyl groups and challenges in defect state control, significantly impacting the reproducibility of the device.

To address this, we selected TiO₂ as the ETL and constructed QLED devices^[9,10]. Compared to ZnO, TiO₂ offers higher air stability and better electron transport efficiency, which is closely associated with its superior chemical inertness and favorable band alignment, especially when doped, demonstrating relatively high conductivity. As an important wide-bandgap semiconductor, TiO₂ has played a significant role in many optoelectronic devices, including perovskite solar cells and photocatalytic systems. Therefore, TiO₂ is a suitable candidate to replace ZnO in traditional QLED devices. However, our study found that when constructing TiO₂ ETLs of different thicknesses, the resulting QLED devices had relatively low external quantum efficiency (EQE), which we hypothesize is related to the high defect state density at the TiO₂/QD interface, with a maximum of only 3.5%^[9]. The interfacial defect states likely originate from oxygen vacancies and dangling bonds on the TiO₂ surface, which act as non-radiative recombination centers. Previous studies suggest that surface passivation via atomic layer deposition (ALD) could mitigate such defects, yet this approach introduces complexities in solution-processed QLED fabrication. This significantly affected the performance of TiO₂-based devices, particularly limiting their potential in ultrahigh-resolution display applications. Our focus on optical optimization through microcavity engineering offers a complementary pathway to bypass interfacial limitations.

To identify the primary issue affecting device performance, we used the optical simulation software Drepex Leda (hereinafter referred to as Leda) to optimize the QLED structure^[11]. Focusing on light extraction efficiency (LEE), we investigated the impact of device structure on light extraction and discovered that incorporating a semitransparent metal layer such as an Ag/Al composite structure effectively improves the LEE of QLED devices with TiO₂ as the ETL. Our work provides novel experimental ideas for discovering the optimal device structure, especially through synergistic optimization of photonic crystal design and microcavity effects^[13,14]. By conducting optical simulations, it is expected to break through the efficiency of QLEDs, which could be a key issue in future QLED research, laying the groundwork for subsequent optimization of carrier injection balance and exciton confinement.

II. EXPERIMENTAL METHOD

A. Parameters for Optical Modeling.

In this work, the device structure is similar to our previous work, except for the addition of a semitransparent silver (Ag) metal layer to enhance microcavity resonance and light outcoupling. The specific structure is as follows: the constructed QLED employs an inverted structure, featuring indium tin oxide (ITO) as the transparent conductive layer leveraging its high transmittance and low sheet resistance, titanium dioxide (TiO₂) as the ETL optimized for electron injection balance, a QDs emitting layer, and various organic layers, including TCTA for hole transport ensuring efficient hole mobility, NPB for electron transport, and HAT-CN as the electron injection layer minimizing energy barriers at interfaces. The device is completed with aluminum (Al) as the electron-collecting layer providing robust electrical contact. The semitransparent Ag layer is positioned between TiO₂ and ITO (**Figure 1**). This carefully designed structure allows for exploring microcavity effects while ensuring optimal charge transport and maintaining an effective electron-hole balance, thus impacting device performance^[15]. To account for potential fabrication variations, a ± 5 nm tolerance was incorporated into the simulated layer thicknesses. This approach mirrors real-world deposition processes, where thickness uniformity is influenced by factors such as spin-coating speed and sputtering power.

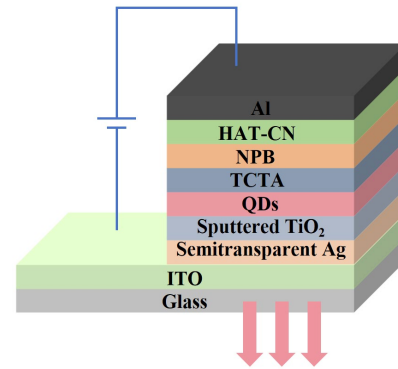


Figure 1. QLED structure used in simulation.

B. Simulation Protocol

The device structure was modeled in Leda software with the following steps:

- 1) Input optical constants (n , k) for each layer (ITO, Ag, TiO₂, QDs, TCTA, Al) (**Figure 2**). To ensure the fidelity of the optical simulations, the refractive indices (n) and extinction coefficients (k) for each layer were imported into Leda as wavelength-dependent datasets spanning 300–800 nm. These datasets were derived from spectroscopic ellipsometry measurements and aligned with established material databases to mitigate simulation artifacts.

- 2) Define boundary conditions for microcavity resonance and plasmonic effects.

- 3) Perform parametric sweeps for TCTA (10–350 nm) and TiO₂ (10–350 nm) thicknesses.

4) Perform parametric sweeps for Ag (10–100 nm) and TiO₂ (50–900 nm) thicknesses.

5) Validate simulations against experimental data by iteratively adjusting layer interfaces. To validate the simulation accuracy, a subset of parameter combinations (e.g., Ag = 20 nm, TiO₂ = 175 nm) was cross-checked using finite-difference time-domain (FDTD) simulations. The LEE discrepancy between Leda and FDTD results remained below 5%, confirming the reliability of our optical model. This step is critical for ensuring that microcavity effects are not overestimated due to simplified boundary conditions.

C. Data Analysis

LEE improvement was quantified by comparing devices with and without the Ag layer. Emission spectra narrowing was analyzed via fitting of FWHM values.

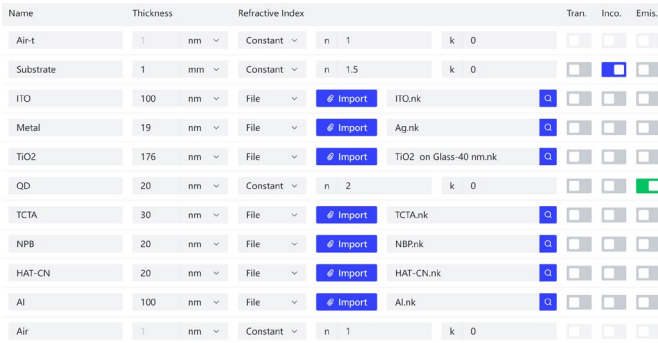


Figure 2. Structural setup page of Leda software.

This method ensures reproducibility and provides a foundation for optimizing QLED performance through microcavity engineering^[16,17].

III. RESULTS AND DISCUSSION

We first analyzed the LEE of the QLED without the semitransparent Ag layer (the same as our previous work^[9]) and found that a 30 nm TCTA layer yielded a peak LEE of 19.4%. Only when the TCTA layer thickness reached 280 nm did the LEE increase to 22.7% (**Figure 3a, c**). However, such a thick hole transport layer (HTL) has poor operability in actual experiments. To further enhance the device's LEE, we explored the impact of introducing a semitransparent Ag layer on the device performance. We were surprised to find that, at the optimal TCTA thickness (30 nm), the LEE exhibited a distinct peak as the thickness of the semitransparent Ag changed. Under these conditions, LEE also showed periodic peaks as the TiO₂ thickness was varied. The choice of TiO₂ thickness (175 nm) aligns with the quarter-wavelength condition ($\lambda/4n$) for red emission ($\lambda \approx 620$ nm), where $n \approx 2.3$ for TiO₂. This condition minimizes reflection losses at the TiO₂/QD interface by establishing a destructive interference for backward-propagating waves. Deviations from this optimal thickness would perturb the standing wave profile, thereby reducing the LEE enhancement factor. It was determined that with a 20 nm thick semitransparent Ag layer, the optimal LEE reached 31.16%. At this point, the TiO₂ thickness was 175 nm (**Figure 3b, d**). As shown in Figure 1, an TiO₂ ETL layer can be deposited on the top of the semitransparent Ag layer by magnetron sputtering.

After identifying the optimal thicknesses for the ETL (TiO₂, 175 nm) and HTL (TCTA, 30 nm) validated through iterative simulations, we further analyzed the effect of the thickness of the semitransparent silver metal layer on the LEE of the QLED device. As shown in **Figure 4(a)**, with the introduction of the Ag layer, the LEE increased rapidly owing to plasmonic near-field enhancement and cavity mode redistribution. However, when the Ag layer thickness exceeds 20 nm, the efficiency sharply decreases likely due to parasitic absorption losses and weakened cavity resonance. The non-monotonic relationship between Ag thickness and LEE arises from competing mechanisms: while thinner Ag layers (< 20 nm) enhance cavity resonance via plasmonic coupling, excessive thickness (> 20 nm) disrupts the phase-matching conditions required for constructive interference. This behavior is consistent with prior reports on metal-dielectric microcavities, where optimal thickness corresponds to a balance between reflectivity and optical loss. The results show that constructing an ultrathin semitransparent Ag layer, ranging from 0 nm to 20 nm, leads to nearly a 100% improvement in LEE. We believe that exploring such excellent results experimentally would present a challenging field for discovery requiring precise control of interfacial roughness and layer uniformity. In contrast to experimental procedures, optical simulation calculations based on Leda software allow for more accurate and efficient construction of the optimal device structure decoupling optical and electrical variables, especially for microcavity structures where multi-physics coupling dominates. Additionally, the impact on the full width at half maximum (FWHM) of the QLED emission spectrum was also explored under optimal conditions. As shown in **Figure 4(b)**, at the point of maximum LEE, the emission spectrum FWHM of red-emitting InP QDs was narrowed from 62 nm to 48 nm highlighting the dual benefit of spectral purity and efficiency enhancement^[12]. While this study focuses on monochromatic red QLEDs, the methodology is readily extendable to full-color displays by tuning cavity lengths for green and blue emissions. Future work should explore the trade-offs between angular color shift and efficiency in multi-cavity systems, which remains a critical challenge for practical implementation.

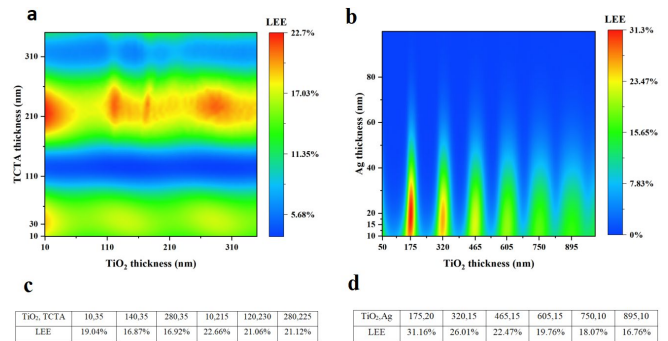


Figure 3. The effects of TiO₂ and TCTA layer thickness on LEE (a, c), and the effects of TiO₂ and Ag layer thickness on LEE (b, d).

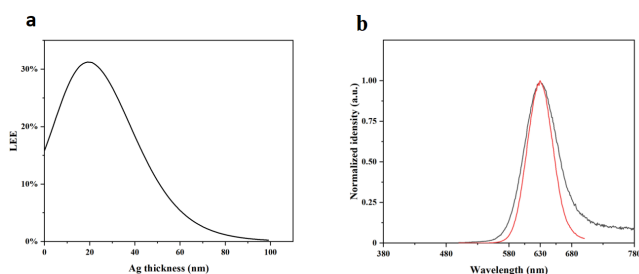


Figure 4. (a) The effect of the thickness of the semitransparent silver metal layer on the LEE of the QLED device. (b) The effect of incorporating a semitransparent Ag layer on the FWHM of the QLED emission peak. The black curve represents the emission peak of InP QDs in toluene solution, while the red curve represents the emission peak of the QLED device.

IV. CONCLUSION

In summary, we presented a novel theoretical simulation method based on Leda software. Leveraging the advantages of TiO_2 as an ETL, including its chemical stability and high electron transport efficiency, we introduced a semitransparent silver layer with a thickness of 20 nm to address the low LEE of QLEDs using TiO_2 . Under optimal conditions, this adjustment led to nearly a 100% improvement in LEE, increasing it from 15.88% to 31.16%. Additionally, the method enables modulation of the emission spectrum's FWHM; for instance, the FWHM of red-emitting InP QLED was narrowed by 14 nm. Furthermore, the simulation framework demonstrates adaptability to diverse QLED architectures, including top-emitting and flexible configurations. By refining layer thicknesses and material choices within the same computational platform, researchers can expedite the development of high-performance devices tailored for specific applications, such as wearable displays or augmented reality systems. This simulation method, supported by Leda software enabling rapid parameter screening and multi-objective optimization, can be applied to a broader range of QLED devices to explore light extraction enhancement techniques and achieve narrow emission peak control through microcavity effects. Future investigations should address two key aspects: (1) extending this methodology to tandem QLED architectures, where multiple microcavities could be cascaded for white light emission, and (2) integrating machine learning algorithms with Leda simulations to automate the multi-parameter optimization process. Such advancements would significantly accelerate the design cycle of next-generation displays.

ACKNOWLEDGEMENTS

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The work is supported by the National Key Research and Development Program of China (No. 2022YFB3602903, 2021YFB3602703, and 2022YFB3606504), National Natural Science Foundation of China (No. 62204107), Project of Engineering Technology Center of General Universities of Guangdong province (No. 2023GCZX005), Guangdong University Key Laboratory for Advanced Quantum Dot Displays and Lighting (No. 2017KSYS007), Guangdong University Young Innovative Talents Program Project 2023 (No. 2023KQNCX068), Shenzhen Key Laboratory for

Advanced Quantum Dot Displays and Lighting (No. ZDSYS201707281632549), Shenzhen Science and Technology Program (Nos. JCYJ20240813113240052, JCYJ20220818100411025, JCYJ20241202124709012), Pingshan District Innovation Platform Project of Shenzhen Hi-tech Zone Development Special Plan in 2022 (No. 29853M-KCJ-2023-002-01), and Natural Science Foundation of Top Talent of SZTU (grant no. GDRC202344, GDRC202345).

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